

PHASE 2 – IDEATION & CONCEPT DEVELOPMENT

Smart Electric Vehicle Battery Health Monitoring System

Project Phase 2 Deliverables

Team members:

1. TAMILINI BHARATHI G
2. UMAMAGESHWARI P
3. VIJAYABHARATHI K
4. VINOTHINI C M

1. Brainstorming Report

The brainstorming process focused on improving the safety, reliability, and lifespan of electric vehicle (EV) batteries. The team identified common battery-related problems such as overheating, overcharging, deep discharge, abnormal current, cell imbalance, inaccurate state-of-charge estimation, and lack of real-time warning. The following ideas were considered:

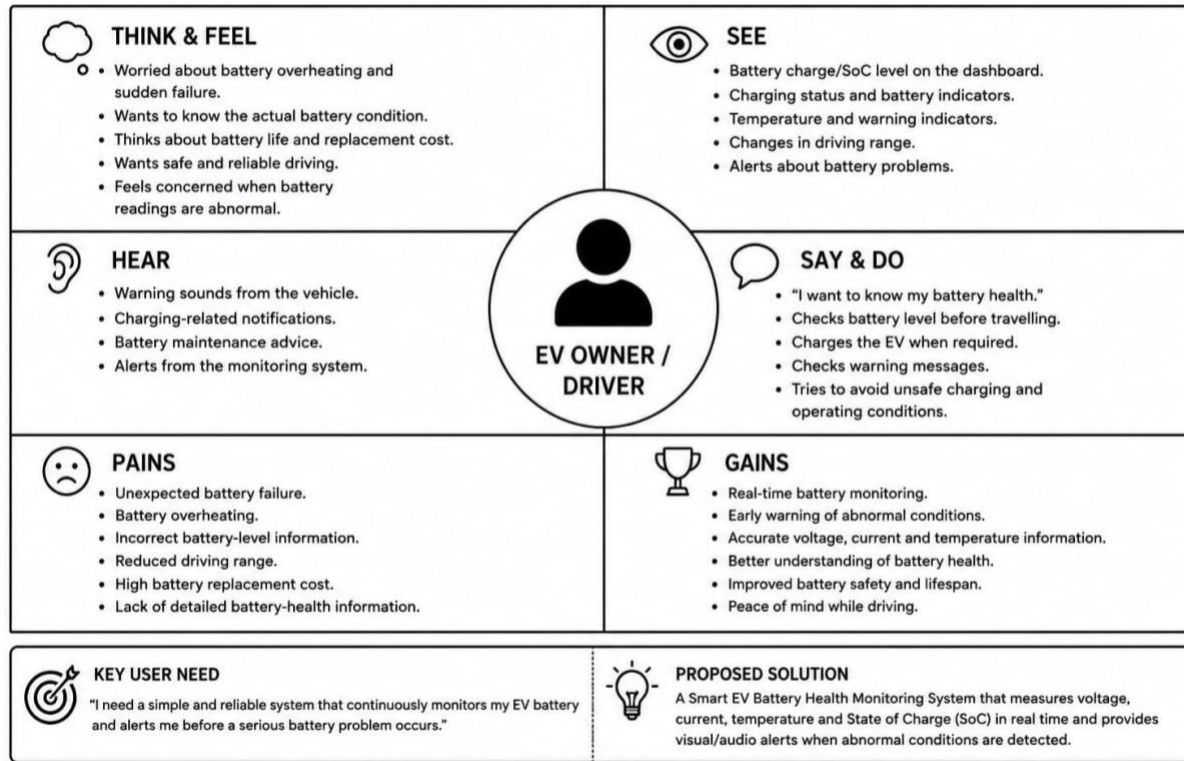
- Real-time voltage and current monitoring
- Battery temperature monitoring and overheating alert
- State of Charge (SoC) estimation
- State of Health (SoH) estimation
- Over-voltage and under-voltage protection
- Over-current and abnormal-current detection
- Cell balancing indication
- IoT/cloud-based remote monitoring
- Mobile/dashboard display for battery parameters
- Emergency warning when unsafe battery conditions are detected

2. Idea Prioritization Matrix

Idea	Safety	Feasibility	Cost	Innovation	Priority
Voltage & current monitoring	High	High	Low	Medium	1
Temperature monitoring	High	High	Low	Medium	2
SoC estimation	High	Medium	Low	High	3
SoH estimation	High	Medium	Medium	High	4
IoT/cloud monitoring	Medium	High	Medium	High	5
Cell balancing	High	Medium	Medium	High	6

Selected concept: A smart battery health monitoring system combining voltage, current, temperature and SoC monitoring, with warning alerts and optional IoT/cloud visualization.

3. Empathy Map



4. Proposed Solution

The proposed Smart Electric Vehicle Battery Health Monitoring System is an embedded monitoring unit that continuously measures important battery parameters. Voltage and current sensors measure electrical conditions, a temperature sensor monitors battery temperature, and a microcontroller processes the sensor data. The system estimates SoC and identifies unsafe operating conditions. A display or IoT dashboard can show real-time information, while a buzzer/notification can provide warnings.

5. Basic functional flow:

EV Battery → Sensors → Microcontroller → Data Processing/SoC Estimation → Display/IoT Dashboard → Warning/Alert

6. Functional Requirements:

S.No	Functional Requirement	Description
1	Continuous Voltage Measurement	Measure the battery voltage continuously to monitor its condition.
2	Current Measurement	Measure the charging and discharging current of the battery.
3	Temperature Measurement	Monitor the battery temperature to prevent overheating.
4	SoC Estimation	Estimate and display the battery State of Charge

		(SoC).
5	Fault Detection	Detect over-voltage, under-voltage, over-current, and abnormal temperature conditions.
6	Warning and Alert	Provide warnings using a buzzer, LED, display, or notification when a fault is detected.
7	Data Storage/Transmission	Store or transmit battery monitoring data when IoT functionality is included.
8	User-Friendly Display	Display voltage, current, temperature, and SoC in a clear and easy-to-understand format.

7. Non-Functional Requirements:

S.No	Non-Functional Requirement	Description
1	Accuracy	Sensor measurements should be sufficiently accurate for reliable battery monitoring.
2	Reliability	The system should operate continuously and provide stable readings.
3	Safety	The monitoring circuit should be properly isolated and protected according to the battery voltage.
4	Low Power Consumption	The monitoring unit should consume minimal energy from the battery.
5	Response Time	Abnormal battery conditions should be detected and alerts should be generated quickly.
6	Usability	Battery readings and alerts should be clear and easy for the user to understand.
7	Scalability	The system should allow additional sensors and IoT features to be added in the future.
8	Maintainability	Sensors and software should be easy to test, troubleshoot, update, and replace.

8. Project Planning – Gantt Chart / Timeline

Activity	W1	W2	W3	W4	W5	W6	W7
Requirement & literature study	✓	✓					
Circuit & component selection		✓	✓				
Sensor interfacing			✓	✓			
Microcontroller programming				✓	✓		
SoC/health monitoring logic					✓	✓	
Prototype integration						✓	✓
Testing & documentation							✓

9. Risk Analysis

Risk	Possible Cause	Impact	Risk Level	Mitigation
Incorrect voltage reading	Sensor/calibration error	Wrong battery status	Medium	Calibrate sensor and verify with a multimeter
Temperature sensor failure	Loose connection/sensor fault	Missed overheating warning	High	Secure sensor and add fault checking
Current measurement error	Noise or sensor offset	Incorrect SoC estimate	Medium	Filtering and calibration
Microcontroller reset	Power instability/software fault	Temporary data loss	Medium	Stable power supply and watchdog
False alarm	Noisy/incorrect threshold	User inconvenience	Medium	Filtering and suitable thresholds
Communication failure	Wi-Fi/Bluetooth loss	Remote data unavailable	Low/Medium	Keep local display/alerts as backup
Battery electrical hazard	Improper wiring/short circuit	Equipment damage/injury	High	Use proper protection, fuses, insulation and supervised testing